

1840s State Debt Credibility — Progress Since Last Meeting

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Research question: did bond markets treat the federal government's refusal to bail out defaulting states in the early 1840s as a credible commitment? The test is whether states that actually defaulted show persistently higher bond yields after the federal no-bailout signal than states that never defaulted, with states that were genuinely at risk but ultimately survived landing in between. It's been about a month since we last met — this document is meant to stand on its own as a full walkthrough of everything done since then, not to assume we're picking up where we left off.

Agenda

Today's flow

- Recap of what was investigated since last meeting
- City-contagion findings (Hall's "disease" question)
- Canal/robustness comparison + Indiana tranche finding
- Anchor-date upgrade
- Two open questions needing a decision
- Proposed structure for the paper, if we're ready to move to writing

What We Investigated Since Last Meeting

Bank-held vs. traded bonds investigation

- You flagged that state-chartered banks were sometimes required to hold state bonds as backing for their own bank notes — a bond held this way would show sparse market data for reasons that have nothing to do with investor sentiment. Built a trade-density flag across all 23 primary/secondary bond codes and checked the two already-known sparse bonds plus the two new ones the flag turned up.
- S-1750 (New York canal, 0.36 obs/month, a 1,666-day gap) is the best circumstantial fit: New York's 1838 Free Banking Act required banks to buy and deposit state bonds against note issuance, and the timing lines up (gap starts 1843, five years after the Act).
- S-2010 (Ohio, 0.60 obs/month) leans toward a different explanation — liquidity concentrating onto a single benchmark issue (S-2080) rather than bank absorption. Ohio's comparable law (the 1845 Kelley Act) postdates the start of this gap by about two years, a timing mismatch.
- Two remain unexplained: S-0040 (Alabama, no maturity data in the codebook at all, only a weak circumstantial link) and S-2190 (Ohio canal, just 2 observations, both from 1825 — a data-coverage limit before the study window even opens, not a trading-pattern finding).
- Bottom line: mixed, honest result — not all four sparse bonds share one cause. Doesn't change any existing chart (none of the four are used as a state's sole evidence anywhere), but it's a real caveat on data quality for the still-secondary canal comparison.

Alabama railroad bond flag

- A source we ran into mentioned Alabama paying bondholders roughly 30 cents on the dollar on railroad-guaranteed bonds — unclear at the time whether that was the same 1840s episode.

- Resolved: it's a distinct, later episode. Alabama's 1867 internal-improvements act had the governor endorse railroad bonds; the Alabama & Chattanooga Railroad defaulted in Jan 1871; Governor Houston's 1875-76 debt commission recognized \$18M of claims, later cut to \$12.5M, against original face claims around \$30M — consistent with the "30 cents on the dollar" figure, but for Reconstruction-era railroad-aid bonds, not our 1840s state bonds.
- No possible overlap with our data: Alabama's price series (S-0030/S-0040) runs 1842-1853; the railroad bonds weren't issued until 1867, 14 years after our data ends. Confirmed from the same Wallis NBER paper already in our reading list, which separates the two episodes in two distinct passages. Does not touch or complicate the Alabama reclassification question.

Ohio yield-rise claim

- You'd mentioned Ohio (never defaulted, our "safe" bucket) saw its yield rise during the 1840s — flagged as possible evidence contagion reached even the safe bucket. Checked at monthly resolution across all four Ohio primary bonds (S-2100/S-2110/S-2080/S-2010).
- Resolved: real move, wrong window. All four bonds show a synchronized price collapse from ~90 (Oct 1841) to ~48-52, bottoming Mar 12-Apr 9 1842 — peak yields S-2100 16.02% (Mar 26 1842), S-2010 15.85% (Apr 9 1842), S-2110 12.83% (Mar 12 1842), S-2080 11.48% (Mar 12 1842).
- This is the 1837 panic window, not the 1843 policy window: the spike bottoms about 4.5 months before PA's actual default and about 11 months before the Feb 11 1843 no-bailout signal. Yields normalize back to a 6-7% baseline starting May 1843 — the opposite of what policy-driven contagion into the safe bucket would predict.
- Reads as a conflation of the panic window and the policy window — exactly the distinction this project has guarded against from the start. Useful confirmation that the panic/policy split is doing real analytical work, not just bookkeeping.

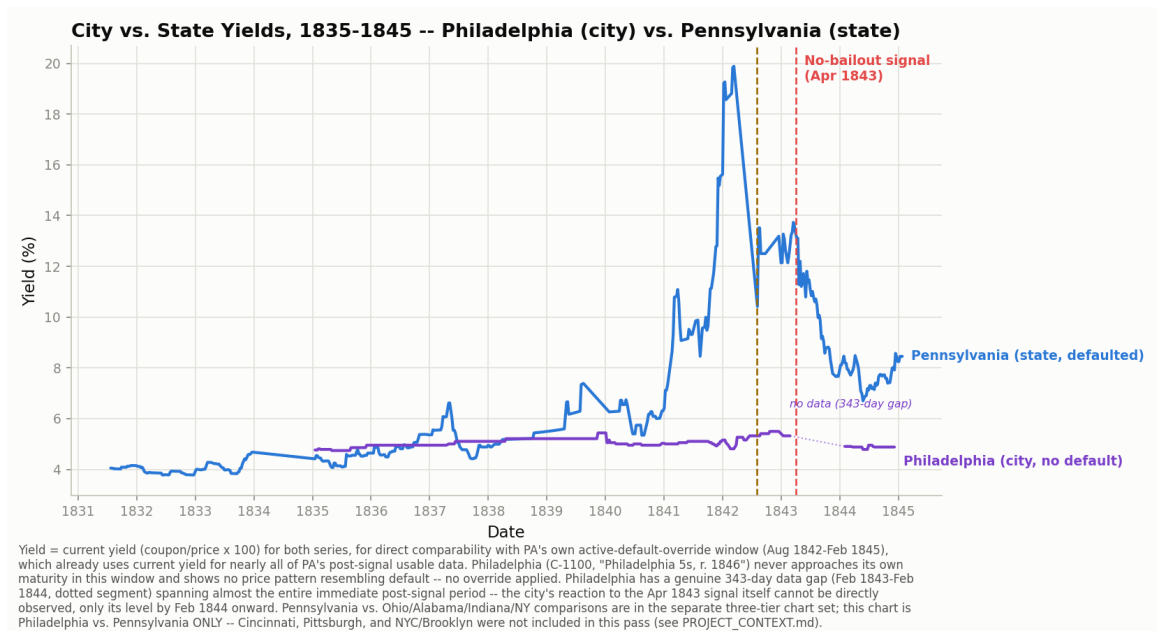
City Contagion — The "Disease" Question

Hall's framing: does state default spread like a disease to the surrounding area — specifically, does it spread downward to the defaulting state's own city? (One clarification: the original notes said "Pennsylvania," but the actual question is Philadelphia — PA's own city — versus Pennsylvania the state, not Pennsylvania versus some other state. That was a note-taking slip we caught and corrected.)

Philadelphia vs. Pennsylvania (C-1100)

- Pre-signal: 1.63pp gap (Pennsylvania 6.69%, Philadelphia 5.06%).
- Post-gap-overlap window (the only period both series actually have data, apples-to-apples): 2.68pp gap (Pennsylvania 7.56%, Philadelphia 4.88%).

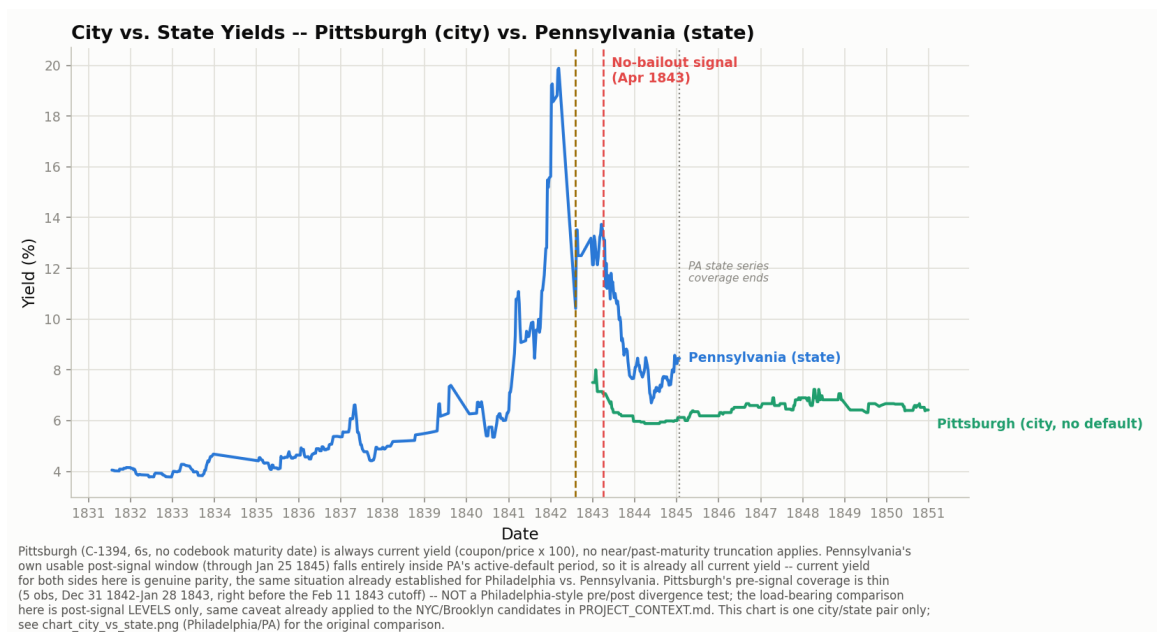
- The widening is driven entirely by Pennsylvania's own yield rising — Philadelphia's yield barely moves at all. The state spikes toward ~20% yield around its own default; Philadelphia stays flat around 5%.



Philadelphia (city) vs. Pennsylvania (state) yields, 1839-1844.

Pittsburgh vs. Pennsylvania

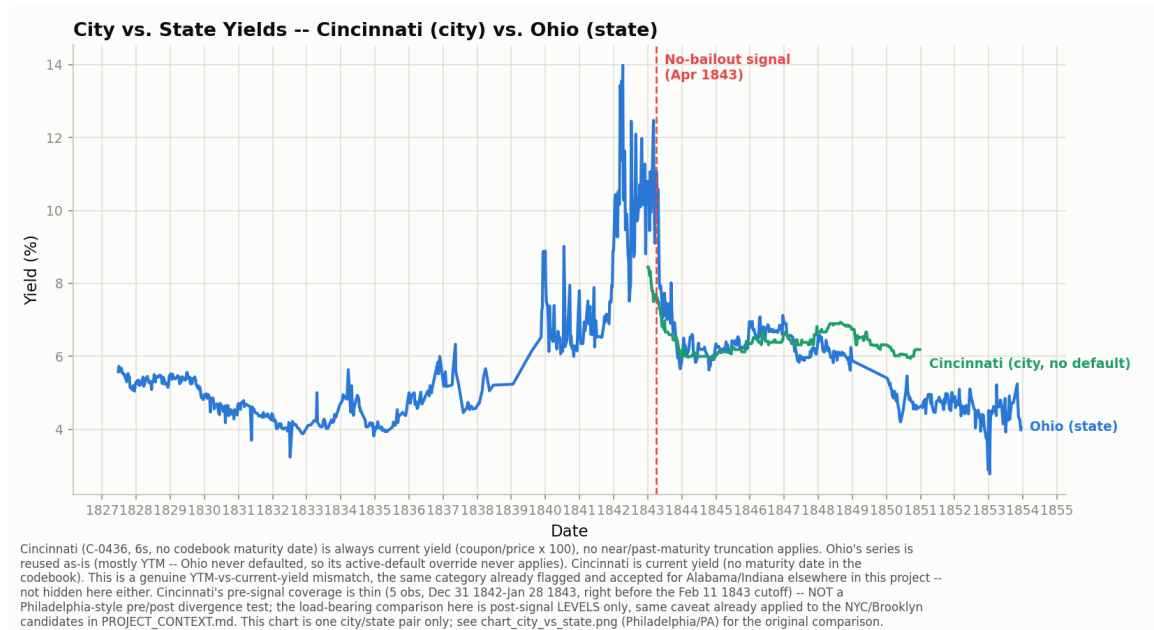
- A second Pennsylvania city, checked to see if Philadelphia's result replicates: it does, almost exactly — 2.65pp gap (Pennsylvania 8.76%, Pittsburgh 6.11%), essentially identical in size and direction to Philadelphia's own 2.68pp.



Pittsburgh (city) vs. Pennsylvania (state) yields.

Cincinnati vs. Ohio — negative control

- Cincinnati vs. Ohio comes in near-parity: -0.11pp (Ohio 6.25%, Cincinnati 6.36%) — essentially flat.
- This is NOT a third confirmation of no-spillover — Ohio never defaulted, so there's no state distress for Cincinnati to be insulated from in the first place. It's a baseline contrast: when the state's own credit is fine, its city tracks it with no gap at all. The ~2.6-2.7pp gap only opens up for the two Pennsylvania cities, where the state itself was genuinely under stress.



Cincinnati (city) vs. Ohio (state) yields — negative control (Ohio never defaulted).

Second Philadelphia bond (C-1260)

- C-1100 has a 343-day data gap (Feb 1843-Feb 1844) that swallows almost the entire immediate post-signal reaction period. A second Philadelphia city bond, C-1260, has real data exactly where C-1100 doesn't.
- Two figures reported together, since they trade off precision against coverage: 5.95pp (Philadelphia 5.59% vs. Pennsylvania 11.53%, n=19, Apr-Aug 1843) using the project's standard post-signal window, or 6.28pp (Philadelphia 5.63% vs. Pennsylvania 11.91%, n=24) using the full window that fills the gap (Feb 25-Aug 5 1843).
- This confirms the state/city gap holds even during the immediate post-signal reaction window, not just before and after it — closing the biggest caveat on the flagship Philadelphia finding.

Summary

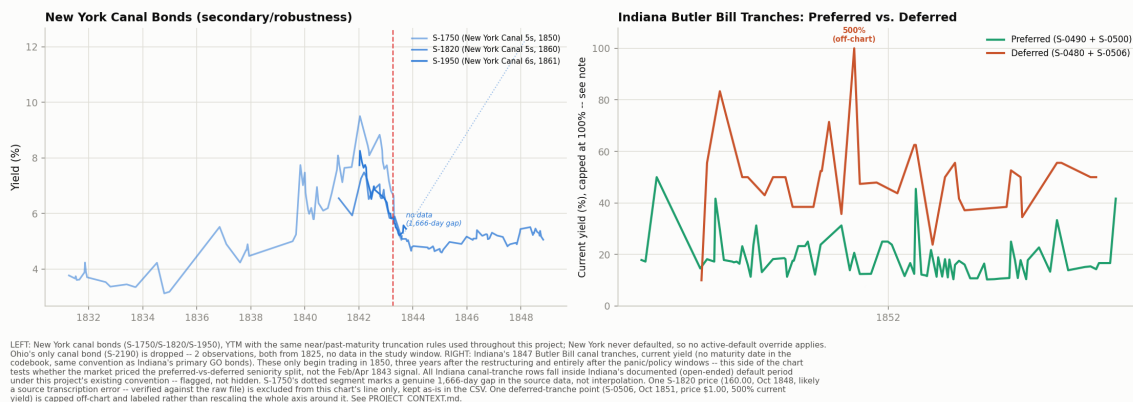
The state gets punished; the city right next to it mostly doesn't — three consistent Pennsylvania-city data points (Philadelphia's C-1100, Philadelphia's C-1260, and Pittsburgh) and one clean negative control (Cincinnati, where the state itself was never under stress).

Canal / Robustness Comparison

A secondary comparison using revenue-pledged canal bonds instead of general-obligation bonds: New York's canal series (S-1750/S-1820/S-1950) and Indiana's 1847 Butler Bill restructuring tranches. Ohio's only canal bond (S-2190) is dropped — 2 observations, both from 1825, no data anywhere in the 1840s-50s study window.

- Headline number: Indiana's 1847 restructuring split its canal debt into a senior "preferred" tranche and a junior "deferred" tranche. The market priced that seniority split by a 40.04pp current-yield gap (preferred 17.96% vs. deferred 57.99%) — the largest spread in the whole project, confirmed directly in raw prices (preferred trades 2-4x deferred throughout), sustained across 146 observations, not a thin-sample artifact.
- Two data-integrity catches made along the way, flagged rather than silently patched: S-1750's known 1,666-day gap is rendered as a dotted, non-connecting chart segment, not a straight line implying a climb that never happened; an isolated S-1820 price of 160.00 (genuine in the raw source, almost certainly a transcription error) is excluded from the chart line but kept as-is in the underlying CSV.

Canal / Revenue-Pledged Bond Robustness Comparison (secondary series)



Canal/robustness comparison: NY canal bonds and Indiana preferred vs. deferred tranches.

Anchor Date Upgrade

- The Feb 11 1843 no-bailout signal date now rests on a real primary source, not just secondary accounts: UNT's digitized Congressional Globe, Vol. 12, 27th Congress, 3rd session, pages 283-284.

- Content: a genuine House floor debate — "Mississippi State Bonds" — with Gwin (MS), Granger (Whig-NY), and Thompson (MS) disputing state bond authorization and federal responsibility for recognizing the debt.
- The exact day within the multi-day debate isn't pinned — neither page carries a visible date stamp — but content, participants, and page position (8-9 pages before an already-confirmed Feb 16 1843 page) all line up tightly.
- Upgrade: from "two secondary sources only" to "secondary sources plus a real, on-topic primary-source read of the right participants and subject in the right tight date window."

Two Things We Need Your Judgment On

1. Alabama reclassification (Defaulted → Risky but survived)

What we know:

- Three independent sources now support it: Wallis's NBER paper explicitly excludes Alabama from the defaulters; two independent lists of 1840s defaulting states both exclude Alabama.

What we need from you:

- A yes/no — or explicit permission to treat this as open in the paper. Not personally confirmed by either of you yet.

2. Indiana's deferred tranche coupon status

What we know:

- Could not determine from available literature whether the deferred tranche stopped paying coupon entirely or just sat lower in priority. Doesn't change the 40.04pp price-based finding — only how precisely it should be worded.

What we need from you:

- Any guidance from the literature — or confirmation it's fine to note as unresolved in the paper.

Also worth flagging, not requiring approval: Cincinnati and the second Philadelphia bond (C-1260) are the newest, least-reviewed pieces of the analysis.

Proposed Paper Structure

If we're ready to start writing — proposed here to confirm or adjust live, not a locked plan.

1. Introduction — the research question, tied to the Lustig "Blue Bonds for Europe" framing.
2. Historical background — 1837 panic, 1840s state defaults, federal refusal to assume debt.

3. Data and methodology — EH.net securities prices, three-bucket classification, panic-window vs. policy-window distinction, active-default override rule.
4. Main result — does the no-bailout signal show up in state-level yields: PA/Ohio/Alabama/Indiana/NY comparison.
5. City contagion result — does default punish the defaulting state's own city (Philadelphia/Pittsburgh/Cincinnati).
6. Robustness — canal bonds, Indiana tranche split, reproducibility checks.
7. Discussion — markets doing real credit analysis on separate legal/fiscal entities rather than generic panic; tie back to the EU "blue bonds" parallel.
8. Conclusion, limitations — NYC/Brooklyn data insufficiency, bond seniority confound, single-day anchor-date uncertainty.